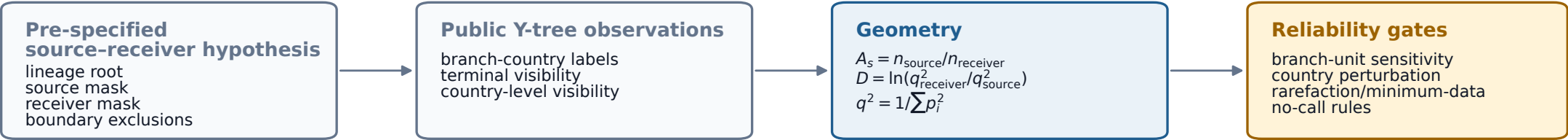


Reliability-aware source-receiver model-checking framework

Pre-specified source-receiver hypotheses in public Y trees are scored as compatibility states, not origin states.

a Workflow for a specified source-receiver hypothesis



Claim discipline: load-bearing, supporting, diagnostic negative, boundary/no-call and unsupported uses remain separated.

b Decision-state matrix

State	Geometry / scoreability	Reliability status	Claim role
● Compatible state	source adequate; receiver branch-depleted	stable	load-bearing eligible
✕ Stable model rejection	specified geometry fails	stable	diagnostic negative
▲ Fragile/support-only state	direction only or scale-sensitive	unstable or bounded	supporting only
■ No-call	insufficient or boundary data	unscoreable	no claim

Boundary: outputs are model-checking states for specified public-tree observations; origin, ethnic identity, continuity, route and ancient presence remain outside the framework's inference.